

back several months; trips to Minnesota to assist his ailing mother (who eventually recovered); and many more, like a broken engagement (though not with Rushdie's ex-girlfriend). "Other than that," it was all within his plan; his own work did not stray the least from schedule. He does not feel responsible for his failure.*

The unexpected has a one-sided effect with projects. Consider the track records of builders, paper writers, and contractors. The unexpected almost always pushes in a single direction: higher costs and a longer time to completion. On very rare occasions, as with the Empire State Building, you get the opposite: shorter completion and lower costs—these occasions are becoming truly exceptional nowadays.

We can run experiments and test for repeatability to verify if such errors in projection are part of human nature. Researchers have tested how students estimate the time needed to complete their projects. In one representative test, they broke a group into two varieties, optimistic and pessimistic. Optimistic students promised twenty-six days; the pessimistic ones forty-seven days. The average actual time to completion turned out to be fifty-six days.

The example of Joe the writer is not acute. I selected it because it concerns a repeatable, routine task—for such tasks our planning errors are milder. With projects of great novelty, such as a military invasion, an all-out war, or something entirely new, errors explode upward. In fact, the more routine the task, the better you learn to forecast. But there is always something nonroutine in our modern environment.

There may be incentives for people to promise shorter completion dates—in order to win the book contract or in order for the builder to get your down payment and use it for his upcoming trip to Antigua. But the planning problem exists even where there is no incentive to underestimate the duration (or the costs) of the task. As I said earlier, we are too narrow-minded a species to consider the possibility of events straying from our mental projections, but furthermore, we are too focused on matters internal to the project to take into account external uncertainty, the "unknown unknown," so to speak, the contents of the unread books.

There is also the nerd effect, which stems from the mental elimination of off-model risks, or *focusing* on what you know. You view the world from *within* a model. Consider that most delays and cost overruns arise from unexpected elements that did not enter into the plan—that is, they lay outside the model at hand—such as strikes, electricity shortages, accidents, bad weather, or rumors of Martian invasions. These small Black Swans that threaten to hamper our projects do not seem to be taken into account. They are too abstract—we don't know how they look and cannot talk about them